

Essential Linux + Git Cheat Sheet

File Management

Command	Description	Example	Explanation
touch	Create empty file	<code>touch file.txt</code>	Creates a blank file or updates its timestamp.
mkdir	Create directory	<code>mkdir myfolder</code>	Creates a new directory with the specified name.
rm	Delete file	<code>rm file.txt</code>	Deletes the specified file permanently.
rmdir	Delete empty directory	<code>rmdir myfolder</code>	Removes a directory only if it's empty.
cp	Copy file or directory	<code>cp file1.txt file2.txt</code>	Copies files or directories from source to destination.
mv	Move or rename	<code>mv old.txt new.txt</code>	Moves or renames a file or directory.
cat	View file content	<code>cat file.txt</code>	Displays the content of a file in the terminal.
stat	File info	<code>stat file.txt</code>	Shows detailed file metadata.
tree	Directory tree	<code>tree /home/user</code>	Displays directory structure in a tree format.
find	Search for files and directories	<code>find / -name file.txt</code>	Searches for files or folders matching criteria.

System Information and Monitoring

Command	Description	Example	Explanation
df / du	Disk usage	<code>df -h, du -sh *</code>	Shows disk space usage for disks and directories.
top / htop	Monitor processes	<code>top</code>	Displays real-time system resource usage.
uptime	Show uptime	<code>uptime</code>	Shows how long the system has been running.
dmesg	Kernel messages	<code>`dmesg</code>	Shows system boot and error messages.
lsof	List open files	<code>lsof -i</code>	Lists open files and associated processes.
env / export	Env variables	<code>export PATH=\$PATH:/newpath</code>	Displays or sets environment variables.

Package Management

Command	Description	Example	Explanation
apt	Debian-based	<code>apt install nginx</code>	Installs packages on Debian systems.
yum / dnf	RHEL-based	<code>yum update</code>	Installs or updates packages on Red Hat systems.

User and Permission Management

Command	Description	Example	Explanation
chmod	Change file permissions	<code>chmod 755 file.sh</code>	Modifies file permissions for user, group, and others.
chown	Change file ownership	<code>chown user:user file.sh</code>	Changes the owner and group of a file.

Networking

Command	Description	Example	Explanation
ip addr	Show IPs	<code>ip addr show</code>	Displays all IP addresses assigned to interfaces.
traceroute	Trace network path	<code>traceroute google.com</code>	Traces the path packets take to reach a host.
nslookup / dig	DNS lookup	<code>nslookup google.com</code>	Queries DNS for resolving domain names.
ping	Check network	<code>ping google.com</code>	Tests connectivity to a host.
wget / curl	Download data	<code>wget URL</code>	Downloads files or data from a URL.
ssh	Remote login	<code>ssh user@host</code>	Logs into a remote machine securely.
scp	Secure copy	<code>scp file.txt user@host:/path/</code>	Copies files securely between hosts.

Process Management

Command	Description	Example	Explanation
ps	Show processes	ps aux	Lists all running processes with details.
kill	Terminate process	kill -9 PID	Sends a signal to stop a process.
systemctl	Manage services	systemctl restart nginx	Controls services on systems using systemd.
journalctl	View system logs	journalctl -u nginx	Displays logs collected by the systemd journal.

Disk Management

Command	Description	Example	Explanation
df	Disk free	df -h	Shows available disk space in human-readable format.
du	Disk usage	du -sh *	Summarizes disk usage of each file/folder.
tar	Archive files	tar -czvf archive.tar.gz dir/	Compresses files into a tarball format.

Text Processing

Command	Description	Example	Explanation
awk	Text processing	awk '{print \$1}' file.txt	Extracts and processes data from text files.
sed	Stream editor	sed 's/old/new/g' file.txt	Edits text in a file using pattern matching.
cut	Remove sections	cut -d':' -f1 /etc/passwd	Extracts fields or sections of each line.
xargs	Build command lines	`cat files.txt`	Converts input into arguments for a command.
grep	Search text	grep 'root' /etc/passwd	Finds lines containing a specified string.

head/ tail	File start/end	tail -n 10 log.txt	Views the beginning or end of a file's contents.
sort	Sort file lines	sort names.txt	Organizes lines of a file alphabetically or numerically.
uniq	Remove duplicate lines	uniq list.txt	Removes repeated lines in a sorted file.

System Configuration and Management

Command	Description	Example	Explanation
crontab	Schedule jobs	crontab -e	Edits the cron table for task automation.
clear	Clear terminal	clear	Clears all previous output from the terminal screen.

Git Version Control

Command	Description	Example	Explanation
<code>git init</code>	Start repo	<code>git init</code>	Initializes a new Git repository.
<code>git clone</code>	Copy repo	<code>git clone URL</code>	Downloads a project from a remote repository.
<code>git status</code>	Check state	<code>git status</code>	Shows the status of files in the working directory.
<code>git add</code>	Stage changes	<code>git add .</code>	Adds changes to the staging area.
<code>git commit</code>	Save changes	<code>git commit -m "message"</code>	Records staged changes to the repo.
<code>git log</code>	History	<code>git log</code>	Displays a list of previous commits.
<code>git diff</code>	View changes	<code>git diff</code>	Shows the differences between commits or working files.
<code>git push/</code> <code>pull</code>	Sync repo	<code>git push origin main</code>	Pushes or pulls updates to/from remote repo.
<code>git branch</code>	Manage branches	<code>git branch new-feature</code>	Creates or lists branches.
<code>git</code> <code>checkout</code>	Switch branch	<code>git checkout main</code>	Switches to a specified branch.
<code>git merge</code>	Merge branches	<code>git merge feature</code>	Merges one branch into another.
<code>git reset</code>	Unstage changes	<code>git reset HEAD file.txt</code>	Removes a file from staging area.
<code>git rm</code>	Remove files	<code>git rm file.txt</code>	Deletes a file from working directory and stages the removal.
<code>git stash</code>	Temporarily save changes	<code>git stash</code>	Stashes current changes to a stack.
<code>git stash</code> <code>pop</code>	Reapply stashed changes	<code>git stash pop</code>	Applies the latest stashed changes and removes them from the stack.
<code>git tag</code>	Create tag	<code>git tag v1.0</code>	Tags a specific commit with a label.
<code>git show</code>	Show commit details	<code>git show <commit></code>	Displays details and diff of a specific commit.
<code>git reflog</code>	Show ref log	<code>git reflog</code>	Shows history of branch HEAD changes.